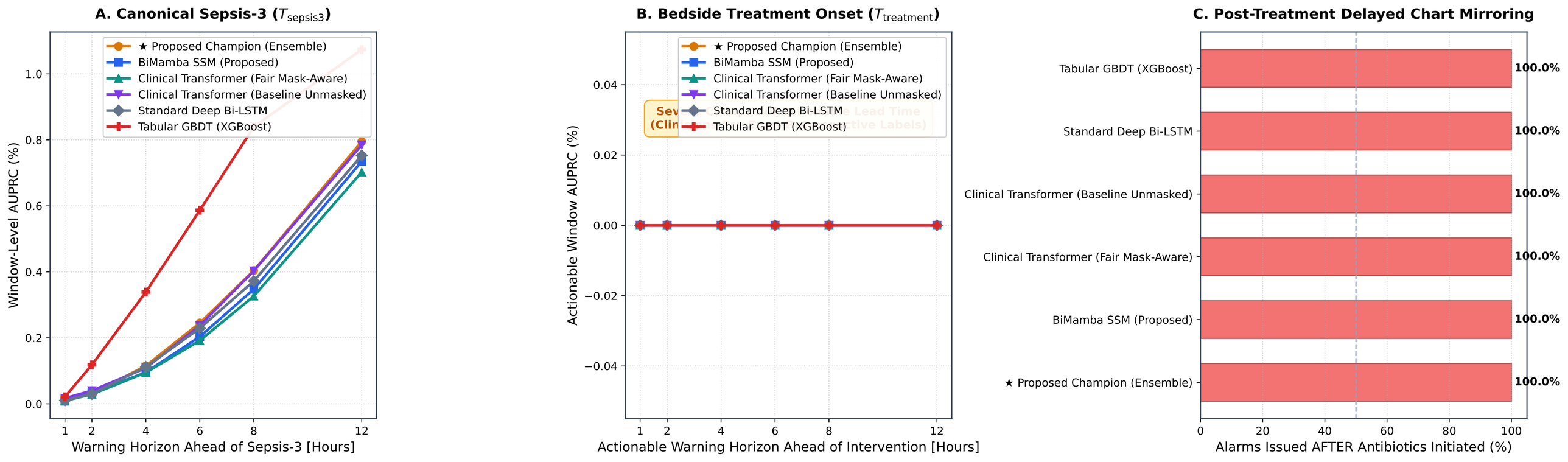
A. Canonical Sepsis-3 ($T_{\text{sepsis3}}$)
★ Proposed Champion (Ensemble)
BiMamba SSM (Proposed)
Clinical Transformer (Fair Mask-Aware)
Clinical Transformer (Baseline Unmasked)
Standard Deep Bi-LSTM
Tabular GBDT (XGBoost)
Window-Level AUPRC (%)
0.0
0.2
0.4
0.6
0.8
1.0
1
2
4
6
8
12
Warning Horizon Ahead of Sepsis-3 [Hours]
B. Bedside Treatment Onset ($T_{\text{treatment}}$)
★ Proposed Champion (Ensemble)
BiMamba SSM (Proposed)
Clinical Transformer (Fair Mask-Aware)
Clinical Transformer (Baseline Unmasked)
Standard Deep Bi-LSTM
Tabular GBDT (XGBoost)
Actionable Window AUPRC (%)
0.04
0.02
0.00
−0.02
−0.04
1
2
4
6
8
12
Actionable Warning Horizon Ahead of Intervention [Hours]
C. Post-Treatment Delayed Chart Mirroring
Tabular GBDT (XGBoost)
Standard Deep Bi-LSTM
Clinical Transformer (Baseline Unmasked)
Clinical Transformer (Fair Mask-Aware)
BiMamba SSM (Proposed)
★ Proposed Champion (Ensemble)
100.0%
100.0%
100.0%
100.0%
100.0%
100.0%
0
20
40
60
80
100
Alarms Issued AFTER Antibiotics Initiated (%)